EXPERIMENT-9

Part-A

AIM: Simulation of Distance Vector Routing (RIP)

OBJECTIVE OF THE EXPERIMENT:

- Understand and implement Distance Vector Routing protocol (RIP v2).
- Observe routing table updates and convergence.
- Compare behavior with static routing.
- Demonstrate periodic updates and multicast behavior.
- Verify end-to-end connectivity using ping and traceroute.

OUTCOME OF THE EXPERIMENT:

Distance Vector Routing calculates distances to every possible destination based on its immediate neighbors. Each router shares its routing table with directly connected routers at regular intervals. This algorithm often employs the **Bellman-Ford algorithm** for route computation. While simple, it has issues like the **Count to Infinity** problem and persistent routing loops.

Description:

Distance Vector Protocols exchange routing tables with their neighbors. RIP is a classic distance-vector protocol using hop count as metric and maximum 15 hops.

RIP v2 (recommended over v1) supports:

- Classless routing (CIDR and subnet masks)
- Multicasting using 224.0.0.9 (instead of broadcast for RIPv2)
- Authentication (plain-text or MD5 in some implementations)

Behavior:

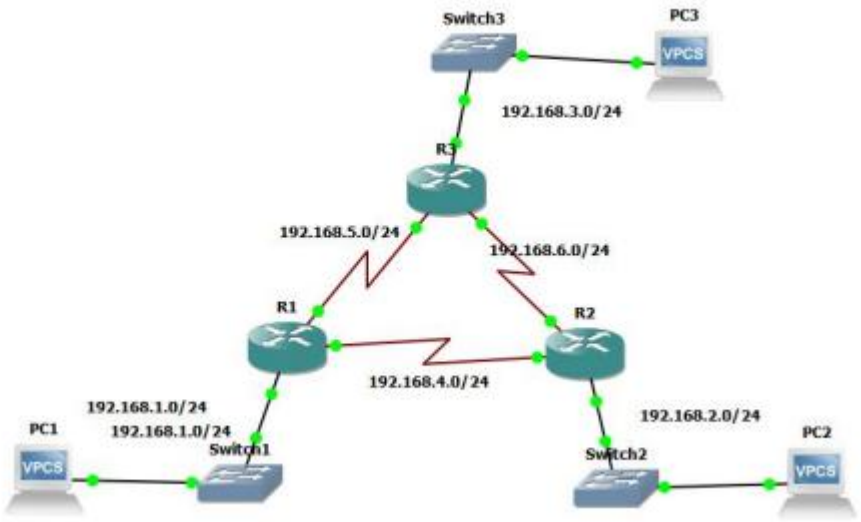
- Sends full routing table every 30 seconds (by default)
- Uses split horizon, route poisoning, triggered updates to mitigate loops but still slower convergence than OSPF

Broadcasting vs Multicasting (RIP)

- RIPv1 used broadcast (255.255.255.255) — inefficient and classful.

RIPv2 uses multicast address 224.0.0.9 — more efficient because only RIP-enabled devices process packets. However, updates are still full-table and periodic.

Topology:



IP Addressing Table:

Device	Interface	IP Address	Subnet mask	Default gateway
PC1	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC2	NIC	192.168.2.2	255.255.255.0	192.168.2.1
PC3	NIC	192.168.3.2	255.255.255.0	192.168.3.1
R1	F0/0	192.168.1.1	255.255.255.0	-
	S1/0	192.168.4.1	255.255.255.0	-
	S1/1	192.168.5.1	255.255.255.0	-
R2	F0/0	192.168.2.1	255.255.255.0	-
	S1/0	192.168.4.2	255.255.255.0	-
	S1/1	192.168.6.1	255.255.255.0	-
R3	F0/0	192.168.3.1	255.255.255.0	-
	S1/0	192.168.5.2	255.255.255.0	-
	S1/1	192.168.6.2	255.255.255.0	-

PSEUDOCODE:

Basic Switch Configuration

Switch> enable
Switch# configure terminal
Switch(config)# hostname SW1
Switch(config)# no ip domain-lookup
Repeat for SW2, SW3

Configure the interfaces on the R1, R2, and R3 routers with the IP addresses from the table under the Topology Diagram.

Configuring interfaces on R1

R1> enable
R1# configure terminal
R1(config)# hostname R1

R1(config)# no ip domain-lookup
R1(config)#interface fastEthernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#interface serial 1/0
R1(config-if)#ip address 192.168.4.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#interface serial 1/1
R1(config-if)#ip address 192.168.5.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
Use the show ip interface brief command to verify that the IP addressing is correct and
that the interfaces are active.
R1# show ip interface brief

Configuring interfaces on R2
R2> enable
R2# configure terminal
R2(config)# hostname R2
R2(config)# no ip domain-lookup
R2(config)#interface fastEthernet 0/0
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface serial 1/0
R2(config-if)#ip address 192.168.4.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface serial 1/1
R2(config-if)#ip address 192.168.6.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#exit

Use the show ip interface brief command to verify that the IP addressing is correct and
that the interfaces are active.
R2# show ip interface brief

Configuring interfaces on R3

R3> enable
R3# configure terminal
R3(config)# hostname R3
R3(config)# no ip domain-lookup
R3(config)#interface fastEthernet 0/0
R3(config-if)#ip address 192.168.3.1 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#interface serial 1/0
R3(config-if)#ip address 192.168.5.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#interface serial 1/1
R3(config-if)#ip address 192.168.6.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#exit

Use the show ip interface brief command to verify that the IP addressing is correct and
that the interfaces are active.
R3# show ip interface brief

Configure the Ethernet interfaces of PC1, PC2, and PC3 with the IP addresses and default
gateways from the table under the Topology Diagram.
Refer to the Addressing Table for PC host address information.
PC1>ip 192.168.1.2/24 192.168.1.1
PC2>ip 192.168.2.2/24 192.168.2.1
PC2>ip 192.168.3.2/24 192.168.3.1

RIP v2 Configuration
R1
R1# configure terminal
R1(config)# router rip
R1(config-router)# version 2
R1(config-router)# network 192.168.1.0
R1(config-router)# network 192.168.4.0
R1(config-router)# network 192.168.5.0
R1(config-router)# exit
R1# write memory

R2

```
R2# configure terminal
R2(config)# router rip
R2(config-router)# version 2
R2(config-router)# network 192.168.2.0
R2(config-router)# network 192.168.4.0
R2(config-router)# network 192.168.6.0
R2(config-router)# exit
R2# write memory
```

R3

```
R3# configure terminal
R3(config)# router rip
R3(config-router)# version 2
R3(config-router)# network 192.168.3.0
R3(config-router)# network 192.168.6.0
R3(config-router)# network 192.168.5.0
R3(config-router)# exit
R3# write memory
```

RESULTS:

Verify and Test Network Connectivity.

PC1> ping 192.168.2.2

View routing protocol information.

R1#show ip protocols

View the routing table on the R1 router.

R1#show ip route

Observations

- After RIP configuration, routers automatically learn remote networks.
- Routing table entries appear as R for RIP.
- Periodic updates are multicast (224.0.0.9).
- Connectivity between PCs across different LANs is successful.
- Convergence is slower compared to OSPF; if a link fails, updates take ~30 sec intervals.

Conclusion

RIP v2 successfully demonstrates Distance Vector Routing:

- Automates route propagation.
- Uses hop-count as metric (max 15).

Supports classless networks and multicast updates.

Part-B

AIM: To configure OSPF (Open Shortest Path First) on multiple routers using Cisco Packet Tracer, observe link-state advertisements, verify routing tables, and test end-to-end connectivity between hosts across multiple LANs**.**

OBJECTIVE OUTCOME OF THE EXPERIMENT:

- Understand and implement OSPF (link-state) routing protocol.
- Learn about OSPF areas, DR/BDR roles, and LSAs.
- Compare convergence speed with RIP (distance vector).
- Observe multicast updates and link-state database.
- Verify routing using ping and traceroute.

OUTCOME OF THE EXPERIMENT:

Link State Routing vs Distance Vector Routing

Routing is a crucial process in computer networks that determines the best path for data packets to travel from one node to another. Two prominent dynamic routing algorithms are **Distance Vector Routing** and **Link State Routing**. Both have distinct methods for calculating routes and disseminating information throughout a network.

Description:

Link-State Routing Protocols

- Routers **share link-state information** with all routers in the network.
- Each router independently calculates the **shortest path using Dijkstra's algorithm**.
- Faster convergence compared to distance vector protocols.
- More scalable for large networks.

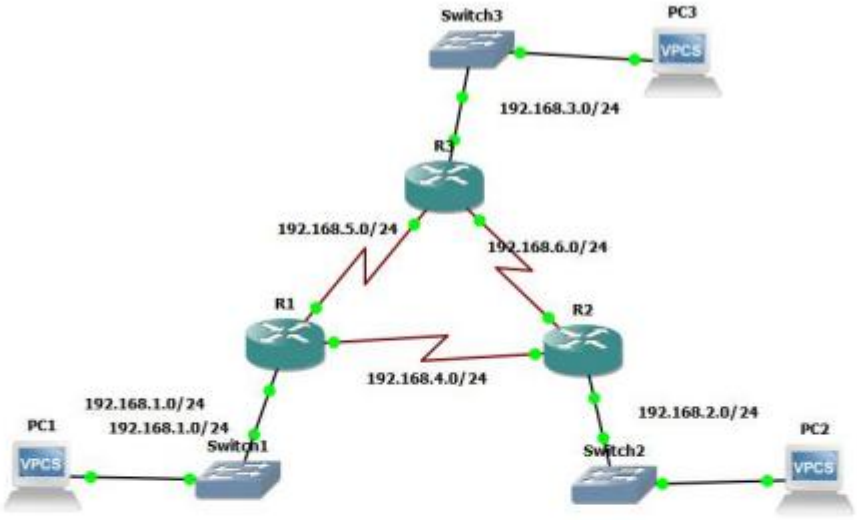
OSPF Features

- Classless, supports VLSM.
- Uses **cost metric** (based on bandwidth).
- Sends updates as **multicast**: 224.0.0.5 (all OSPF routers), 224.0.0.6 (all DR/BDR).
- Supports **areas** to divide networks for scalability.
- **Designated Router (DR) and Backup DR (BDR)** reduce LSA flooding in multi-access networks.

OSPF Areas

- **Area 0**: Backbone area; all other areas must connect to it.
- Other areas (1,2…) reduce routing table size and control flooding.
- Routers in same area share link-state database.

Topology:



IP Addressing Table:

Device	Interface	IP Address	Subnet mask	Default gateway
PC1	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC2	NIC	192.168.2.2	255.255.255.0	192.168.2.1
PC3	NIC	192.168.3.2	255.255.255.0	192.168.3.1
R1	F0/0	192.168.1.1	255.255.255.0	-
	S1/0	192.168.4.1	255.255.255.0	-
	S1/1	192.168.5.1	255.255.255.0	-
R2	F0/0	192.168.2.1	255.255.255.0	-
	S1/0	192.168.4.2	255.255.255.0	-
	S1/1	192.168.6.1	255.255.255.0	-
R3	F0/0	192.168.3.1	255.255.255.0	-
	S1/0	192.168.5.2	255.255.255.0	-
	S1/1	192.168.6.2	255.255.255.0	-

Configure Basic Network Device Settings.

Basic Switch Configuration

Switch> enable
Switch# configure terminal
Switch(config)# hostname SW1
Switch(config)# no ip domain-lookup
Repeat for SW2, SW3

Configure the interfaces on the R1, R2, and R3 routers with the IP addresses from the table under the Topology Diagram.

Configuring interfaces on R1

R1> enable
R1# configure terminal

R1(config)# hostname R1
R1(config)# no ip domain-lookup
R1(config)#interface fastEthernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#interface serial 1/0
R1(config-if)#ip address 192.168.4.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#interface serial 1/1
R1(config-if)#ip address 192.168.5.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
Use the show ip interface brief command to verify that the IP addressing is correct and that the interfaces are active.
R1# show ip interface brief

Configuring interfaces on R2
R2> enable
R2# configure terminal
R2(config)# hostname R2
R2(config)# no ip domain-lookup
R2(config)#interface fastEthernet 0/0
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface serial 1/0
R2(config-if)#ip address 192.168.4.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface serial 1/1
R2(config-if)#ip address 192.168.6.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#exit

Use the show ip interface brief command to verify that the IP addressing is correct and that the interfaces are active.
R2# show ip interface brief

Configuring interfaces on R3
R3> enable

R3# configure terminal
R3(config)# hostname R3
R3(config)# no ip domain-lookup
R3(config)#interface fastEthernet 0/0
R3(config-if)#ip address 192.168.3.1 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#interface serial 1/0
R3(config-if)#ip address 192.168.5.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#interface serial 1/1
R3(config-if)#ip address 192.168.6.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#exit

Use the show ip interface brief command to verify that the IP addressing is correct and that the interfaces are active.
R3# show ip interface brief

Configure the Ethernet interfaces of PC1, PC2, and PC3 with the IP addresses and default gateways from the table under the Topology Diagram.
Refer to the Addressing Table for PC host address information.
PC1>ip 192.168.1.2/24 192.168.1.1
PC2>ip 192.168.2.2/24 192.168.2.1
PC2>ip 192.168.3.2/24 192.168.3.1

Configure OSPF
Configure OSPF on R1
Use the router ospf command in global configuration mode to enable OSPF on the R1 router. Enter a process ID of 1 for the process-ID parameter.
R1(config)#router ospf 1
R1(config-router)#

Once you are in the Router OSPF configuration sub-mode, configure the LAN network. Use an area ID of 0 for the OSPF area-id parameter. 0 will be used for the OSPF area ID in all of the network statements in this topology.

R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# network 192.168.4.0 0.0.0.255 area 0
R1(config-router)# network 192.168.5.0 0.0.0.255 area 0

R1(config-router)#

Configuring OSPF on R2

R2(config)#router ospf 1
R2(config-router)# network 192.168.2.0 0.0.0.255 area 0
R2(config-router)# network 192.168.4.0 0.0.0.255 area 0
R2(config-router)# network 192.168.6.0 0.0.0.255 area 0
R2(config-router)#

Notice that when the network for the serial link from R1 to R2 is added to the OSPF configuration, the router sends a notification message to the console stating that a neighbor relationship with another OSPF router has been established.

Configuring OSPF on R3

R3(config)#router ospf 1
R3(config-router)# network 192.168.3.0 0.0.0.255 area 0
R3(config-router)# network 192.168.5.0 0.0.0.255 area 0
R3(config-router)# network 192.168.6.0 0.0.0.255 area 0
R2(config-router)# exit
R2# write memory

Results:

Verify and Test Network Connectivity.

PC1> ping 192.168.2.2

View routing protocol information.

R1#show ip protocols

View the routing table on the R1 router.

R1#show ip route

Verify ospf Operation- View OSPF neighbors

R1#show ip ospf neighbors

R1#show ip ospf database → Link-State Database (LSAs)

Conclusion

- OSPF successfully demonstrates **link-state routing**.
- Provides **fast convergence**, supports large networks via areas.
- Uses **cost metric** and multicasting for efficiency.
- Preferred for medium-to-large enterprise networks.

Pre Lab Questions:

1. What is routing?
2. What is a routing algorithm?
3. What is Distance Vector Routing?

Sample Viva Voce Questions:

1. What is the main purpose of this experiment?
2. What does "distance" mean in Distance Vector Routing?
3. What does "vector" mean?

Q	A
1. What is RIP?	Routing Information Protocol; a Distance Vector routing protocol.
2. Maximum hop count in RIP?	15 hops; 16 = unreachable.
3. Difference between RIP v1 and v2?	v1: classful, broadcast updates; v2: classless, multicast updates, supports subnet masks.
4. Administrative distance of RIP?	120
5. Why use no auto-summary?	Prevents automatic summarization at classful boundaries.
6. Which multicast address does RIP v2 use?	224.0.0.9
7. How to verify RIP routes?	show ip route (look for R entries).
8. Difference between static and RIP routing?	Static: manual routes, no automatic update; RIP: routes learned dynamically.
9. How often does RIP send updates?	Every 30 seconds (default).
1. What is OSPF?	Open Shortest Path First, a link-state routing protocol.
2. What metric does OSPF use?	Cost based on bandwidth.
3. What is a router ID?	Unique identifier (32-bit) for OSPF router.
4. What are OSPF areas?	Logical grouping of routers to reduce flooding & routing table size.
5. What is DR and BDR?	Designated Router and Backup Designated Router; reduce LSA flooding on LANs.
6. Multicast addresses for OSPF?	224.0.0.5 (all OSPF routers), 224.0.0.6 (DR/BDR).
7. Difference between OSPF and RIP?	OSPF: link-state, fast convergence, scalable; RIP: distance vector, hop count metric, slow convergence.
8. How to verify OSPF neighbors?	show ip ospf neighbor
9. What command shows OSPF routes?	show ip route (look for O)
10. How to save OSPF configuration?	write memory or copy running-config startup-config